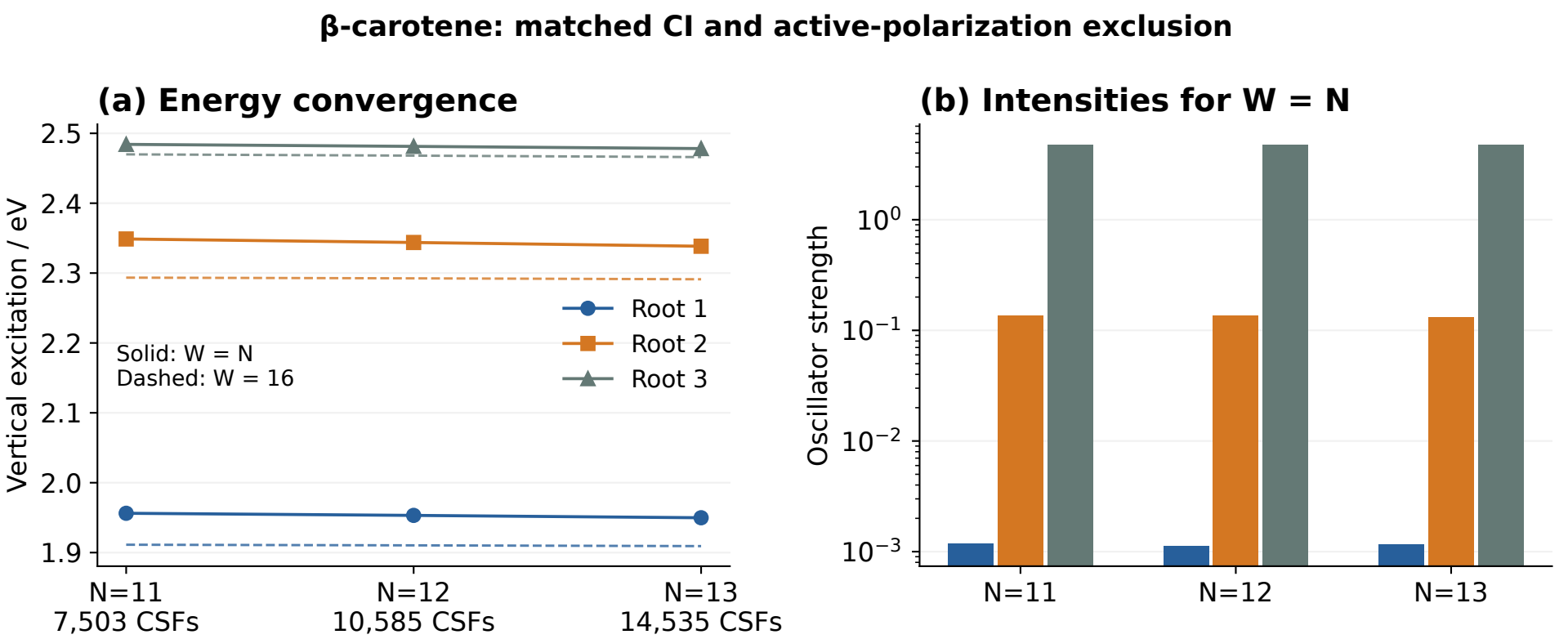
β-carotene: matched CI and active-polarization exclusion
(a) Energy convergence
Vertical excitation / eV
2.5
2.4
2.3
2.2
2.1
2.0
1.9
Solid: W = N
Dashed: W = 16
Root 1
Root 2
Root 3
N=11
7,503 CSFs
N=12
10,585 CSFs
N=13
14,535 CSFs
(b) Intensities for W = N
Oscillator strength
10^{0}
10^{-1}
10^{-2}
10^{-3}
N=11
N=12
N=13